# Alan Mozhoor

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## EDUCATION

**Georgia Institute of Technology, Atlanta, GA** **Expected May 2029**

*Bachelor of Science in Mathematics & Computing* GPA: 4.00/4.00

Concentration: Mathematical Intelligence and Data Science | ACT: 35 (STEM: 36)

**Honors:** MathWorks M3 Challenge Honorable Mention ($1,000 scholarship), Mar 2026

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML/CSS

**Machine Learning & Data:** Reinforcement Learning (SAC, PPO, DQN, HER), Deep Learning, Quantization, Embeddings, Time Series

**APIs & Backend:** REST APIs, FastAPI, Webhooks, PostgreSQL/Supabase, Twilio, OpenAI/Anthropic APIs, Make (Automation)

**Certifications:** Certiport (Python, Java, Software Dev, AI, Data Analytics); Microsoft (Azure Fundamentals, AI Fundamentals)

**Libraries & Tools:** PyTorch, JAX, MuJoCo, Gymnasium, TensorFlow, scikit-learn, LightGBM, NumPy, pandas, Next.js, Git

## EXPERIENCE

**Ascent Math (joinascentmath.com), Remote** | *Co-Founder & Head of Engineering* **May 2026 – Present**

- Architected an event-driven operations platform integrating scheduling, customer lifecycle management, and asynchronous communication workflows; automated 85% of administrative operations while sustaining 95% fulfillment across 150+ sessions
- Engineered API-driven lead ingestion, scheduling orchestration, and automated follow-up pipelines processing 400+ inquiries; achieved 35% conversion and supported 3× month-over-month growth across a 100+ student platform

**The Pert Group LLC, Alpharetta, GA** | *Automation Developer* **Nov 2025 – Aug 2026**

- Architected and deployed an AI voice agent using Python, Twilio, OpenAI API, and Make, processing 150+ interactions/month, automating 85% of inquiries, order intake, and follow-ups, and reducing handling time by 65%
- Engineered an event-driven pipeline integrating 6+ REST APIs and webhooks, transforming unstructured LLM outputs into structured customer/order data with 82.3% accuracy and sub-second routing
- Productionized fault-tolerant automation with validation, exponential-backoff retries, structured logging, and automated recovery, reaching 83.7% autonomous completion and saving 25+ hours/month

**Georgia Southern University, Statesboro, GA** | *Applied Mathematics & Machine Learning Research* **Jun 2025 – Jul 2025**

- Developed and evaluated a TensorFlow CNN for binary dermatological image classification, achieving ~96% validation accuracy distinguishing tinea from circular eczema
- Investigated the mathematical foundations of deep learning, including convolutional operators, linear transformations, and gradient-based optimization; presented technical findings during a university-hosted research program

## PROGRAMMING PROJECTS

**Quantization-Aware RL for Robot Arm Control** **Oct 2026 – Present**

- Implemented SAC with Hindsight Experience Replay and a symmetric uniform quantizer (straight-through estimator, fake-quant layers, PTQ calibration) from scratch in PyTorch to measure how few bits a simulated MuJoCo Fetch arm policy needs
- Reached 100% success over 300 FetchReach-v4 evaluation episodes across 3 seeds (3.15 cm mean final error, 95%+ success within 3,000 steps) with bitwise-reproducible training and CI; QAT sweeps in progress

**Ari AI: Autonomous Cybersecurity & Deep Reinforcement Learning** **Oct 2026 – Present**

- Designed a PyTorch Double DQN defense policy for a partially observable, 10-node cyberattack simulator with 84-dimensional state vectors and 36 discrete defensive actions; reduced simulated breach rate from 35.6% to 5.6% against a rule-based baseline over 90 held-out episodes
- Engineered a typed, permission-scoped multi-agent security pipeline with 7 specialized Red/Blue/verifier roles, deterministic state-machine orchestration, and isolated remediation; validated 18 vulnerabilities through executable HTTP tests, automated patch generation, and independent regression verification

**Constellation: Representation Learning & Personalized Ranking System** **Sep 2026**

- Built a hybrid semantic retrieval and learning-to-rank pipeline using BGE-small embeddings, IDF-weighted interest representations, logistic regression, and LightGBM LambdaRank; improved AUC from 0.73 to 0.82 and NDCG@10 from 0.85 to 0.89 on 1,651 synthetic interaction pairs
- Engineered an explainable multi-objective matching system combining facet-wise cosine similarity, weighted Jaccard overlap, bidirectional goal complementarity, and UMAP/HDBSCAN community discovery; exposed feature-level score attribution and graph-based retrieval through FastAPI endpoints

**Robot Arm Vision Policy on FPGA (team project)** **Sep 2026 – Present**

- Helped assemble and set up the SO-101 arm for a team int8 vision policy with 91% sim success (90% under disturbances, 200 episodes each) that runs on a KR260 FPGA at 8 ms per inference, ~9x faster than the board CPU
- Built the RL teacher pipeline for post-training a SmolVLA vision-language-action model: JAX PPO, success-guided sampling (SGS) of reset states, batched MJX/Warp environments, and camera calibration tools for sim-to-real